5058 Homework2

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1. Average Entropy

Let

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p)$$

(a)

Substitute $p = 1/3$:

$$H\left(\frac{1}{3}\right) = -\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right) \approx 0.9183$$

(b)

If $p \sim U(0,1)$,

$$E[H(p)] = \int_0^1 H(p)\,dp$$

Substitute $H(p)$:

$$E[H(p)] = \int_0^1 [-p\log_2 p - (1-p)\log_2(1-p)]dp$$

Using $\log_2 x = \frac{\ln x}{\ln 2}$:

$$E[H(p)] = \frac{-1}{\ln 2}\int_0^1 [p\ln p + (1-p)\ln(1-p)]dp$$

Since

$$\int_0^1 p\ln p\,dp = -\frac{1}{4}, \quad \int_0^1 (1-p)\ln(1-p)\,dp = -\frac{1}{4}$$

$$E[H(p)] = \frac{-1}{\ln 2}\left(-\frac{1}{2}\right) = \frac{1}{2\ln 2} \approx 0.7213$$

2. Mutual Information for Correlated Normal Distributions

Let

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2\left(0, \begin{pmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{pmatrix}\right)$$

Mutual information:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

For Gaussian variables,

$$H(X) = \frac{1}{2}\log\bigl(2\pi e\sigma^2\bigr)$$

Thus

$$H(X) + H(Y) = \log\bigl(2\pi e\sigma^2\bigr)$$

The joint entropy of a bivariate normal is

$$H(X,Y) = \frac{1}{2}\log\bigl((2\pi e)^2|\Sigma|\bigr)$$

where

$$|\Sigma| = \begin{vmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{vmatrix} = \sigma^4(1-\rho^2)$$

Thus

$$H(X,Y) = \log\bigl(2\pi e\sigma^2\bigr) + \frac{1}{2}\log\bigl(1-\rho^2\bigr)$$

Therefore

$$I(X;Y) = -\frac{1}{2}\log\bigl(1-\rho^2\bigr)$$

Cases

- $\rho = 0 : I(X;Y) = -\frac{1}{2}\log(1) = 0$
- $\rho = 1 : I(X;Y) \to \infty$
- $\rho = -1 : I(X;Y) \to \infty$

3. Channel Capacity

Input alphabet: $X = \{0,1\}$ with $P(X=0) = P(X=1) = 1/2$. Output alphabet: $Y = \{0,1\}$. Channel matrix:

$$\begin{pmatrix} 1-\alpha & \beta \\ \alpha & 1-\beta \end{pmatrix}$$

(a)

Entropy of the input:

$$H(X) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1$$

(b)

Output distribution:

$$\begin{aligned} P(Y=0) &= P(Y=0|X=0)P(X=0) + P(Y=0|X=1)P(X=1) \\ &= \frac{1}{2}(1-\alpha) + \frac{1}{2}\beta = \frac{1-\alpha+\beta}{2} \end{aligned}$$

$$P(Y=1) = \frac{1}{2}\alpha + \frac{1}{2}(1-\beta) = \frac{1+\alpha-\beta}{2}$$

Entropy:

$$H(Y) = -P(Y=0)\log_2 P(Y=0) - P(Y=1)\log_2 P(Y=1)$$

(c)

Joint distribution:

$$\begin{aligned} P(0,0) &= \frac{1}{2}(1-\alpha) \\ P(0,1) &= \frac{1}{2}\alpha \\ P(1,0) &= \frac{1}{2}\beta \\ P(1,1) &= \frac{1}{2}(1-\beta) \end{aligned}$$

Joint entropy:

$$H(X,Y) = -\sum_{x,y} P(x,y)\log_2 P(x,y)$$

(d)

Mutual information:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

Using $H(X) = 1$:

$$I(X;Y) = 1 + H(Y) - H(X,Y)$$

Equivalently,

$$I(X;Y) = H(Y) - H(Y|X)$$

Conditional entropy:

$$H(Y|X) = \frac{1}{2}H(\alpha) + \frac{1}{2}H(\beta)$$

where

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p)$$

Thus

$$I(X;Y) = H\left(\frac{1-\alpha+\beta}{2}\right) - \frac{1}{2}H(\alpha) - \frac{1}{2}H(\beta)$$

(e)

The mutual information is maximal for **two** combinations of (α, β):

$$(\alpha, \beta) = (0, 0) \quad \text{and} \quad (1, 1)$$

In the first case $Y = X$, in the second case $Y = 1 - X$. In both cases the channel is noiseless (deterministic and invertible), thus

$$I(X;Y) = H(X) = 1$$

Channel capacity:

$$C = 1 \text{ bit}$$

(f)

Capacity is minimal when the output is independent of the input. This occurs when

$$\alpha + \beta = 1$$

In this case the two rows of the channel matrix are identical, so Y is statistically independent of X, and

$$I(X;Y) = 0$$

Thus the channel capacity is

$$C = 0$$

(4) Shannon, Fano, and Huffman Codes

A source X has an alphabet $\{a, b, c, d, e, f, g\}$ with probabilities

$$\{0.01,\ 0.24,\ 0.05,\ 0.20,\ 0.47,\ 0.01,\ 0.02\}.$$

(a) Entropy of X

The entropy of a discrete source is

$$H(X) = -\sum_i p(x_i) \log_2 p(x_i)$$

Thus

$$\begin{aligned} H(X) = -\ [&0.01 \log_2 0.01 + 0.24 \log_2 0.24 + 0.05 \log_2 0.05 \\ &+ 0.20 \log_2 0.20 + 0.47 \log_2 0.47 \\ &+ 0.01 \log_2 0.01 + 0.02 \log_2 0.02] \end{aligned}$$

$$H(X) \approx 1.93 \text{ bits}$$

(b) Shannon, Fano and Huffman Codes

First sort symbols by decreasing probability:

$$e(0.47),\ b(0.24),\ d(0.20),\ c(0.05),\ g(0.02),\ a(0.01),\ f(0.01)$$

Shannon code Code lengths are

$$l_i = \lceil -\log_2 p_i \rceil$$

This gives

Symbol	$p(x)$	Code
e	0.47	00
b	0.24	011
d	0.20	101
c	0.05	11101
g	0.02	111101
a	0.01	1111101
f	0.01	1111110

Fano code By recursively splitting the probabilities into two groups with approximately equal total probability, one possible Fano code is

Symbol	Code
e	0
b	10
d	110
c	1110
g	11110
a	111110
f	111111

Huffman code Constructing the Huffman tree gives an optimal prefix code. One possible result is

Symbol	Code
e	0
b	10
d	111
c	1101
g	11001
a	110000
f	110001

(c) Average Code Length

The average code length is

$$L = \sum p(x_i) l_i$$

For the Shannon code

$$L_{Shannon} \approx 2.77 \text{ bits}$$

For the Fano code

$$L_{Fano} \approx 1.97 \text{ bits}$$

For the Huffman code

$$L_{Huffman} \approx 1.97 \text{ bits}$$

(d) Optimal Code

The optimal code is the **Huffman code** because it minimizes the expected codeword length among all prefix codes. The entropy is

$$H(X) \approx 1.93$$

and the Huffman average length is

$$L_{Huffman} \approx 1.97$$

Therefore the difference is

$$L_{Huffman} - H(X) \approx 0.04 \text{ bits}$$

which shows that the Huffman code is very close to the theoretical lower bound given by the entropy.